

SageServe : MLOps Integration Plan

Potential Project Idea

SageServe remains a **simulation-based evaluation framework** for forecast-aware autoscaling.



Note

We will add an **MLOps pipeline around the workload data and demand-forecasting model**, rather than treating the simulator itself as an ML system.

MLOps Components to Implement

Configuration Management

- Make the pipeline configuration-driven.
- Use YAML/Hydra configuration for:
 - trace selection
 - forecasting model
 - forecast horizon/frequency
 - simulator settings
 - evaluation settings

ETL + Data Validation

Build a preprocessing pipeline:



Validate:

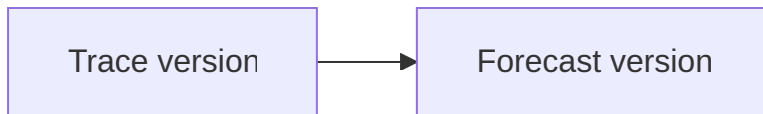
- required columns
- timestamps
- missing values
- duplicate
- records
- negative/invalid workload values
- valid model/region IDs

Versioning + Lineage

Version:

- raw traces
- processed forecasting datasets (generated) forecasts
- experiment/simulation runs

Maintain lineage:



A lightweight Git/DVC-based implementation is sufficient.

Model Registry + Governance

Treat the demand forecaster as the ML model.

Initially use the existing **ARIMA forecasting approach**.

Register:

- forecast model version
- training dataset version
- parameters (MAE/RMSE/MAPE)
- training timestamp
- validation status

The selected model version produces the forecast consumed by SageServe.

Containerization

Containerize the complete reproducible workflow:

```
+ Data preprocessing  
  
+ Forecasting  
  
+ SageServe  
  
+ Evaluation
```

The simulator does not need to become a production service.

CI/CD + Testing

Create a GitHub Actions pipeline that automatically checks:

- data/schema validation
- preprocessing
- forecasting output
- model sanity checks
- SageServe simulator smoke test
- metric generation

Example:



Observability + Drift Monitoring

Monitor the following

Data drift :

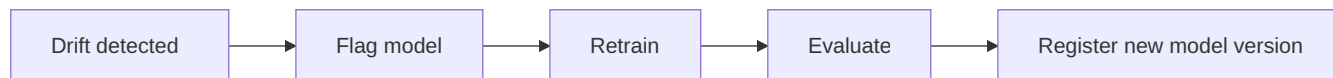
- request rate
- token/request distribution
- model distribution
- regional distribution

- workload type

Model performance :

- predicted vs actual demand
- MAE/RMSE/MAPE

If drift or forecast error exceeds a configured threshold:



SageServe's Role

We will **not** modify SageServe into an ML system.

Instead:



The simulator is the **downstream evaluation environment** for different forecasting model versions.

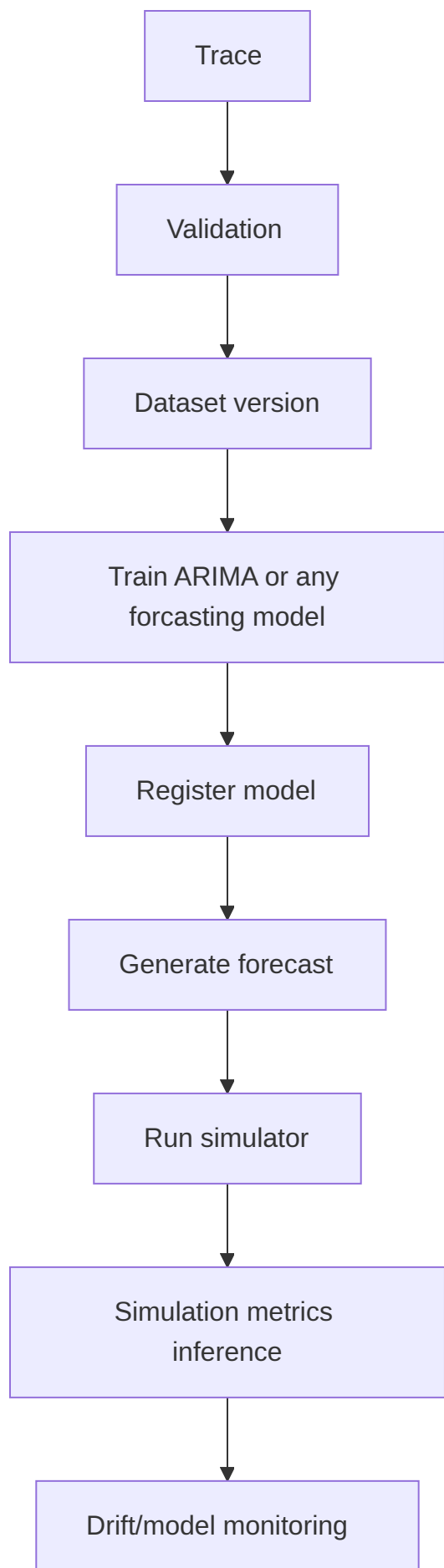
This allows us to answer:

? Question

Does a new forecasting model not only improve prediction accuracy, but also improve the resulting autoscaling decisions and system-level metrics?

Final Demonstration

The final demo should show one complete run:



Then compare **at least two model versions/configurations** through the same SageServe simulation and show both:

1. forecasting metrics, and
2. downstream simulation/autoscaling metrics.

The goal is to demonstrate that MLOps manages the **entire lifecycle of the forecasting component**, while SageServe evaluates its effect on the simulated serving system.
